STUDY SMARTER, NOT HARDER / WHAT REALLY PREDICTS YOUR FINAL GRADE

Who gets the higher grade?

STUDENT 1 · THE GRINDER

10 h of study a week

20 classes missed

0 votes

VS

STUDENT 2 · THE REGULAR

2 h of study a week

0 classes missed

0 votes

Stats details N

Made-up students. Click a card (or press 1 / 2) to count a hand.

Undo last

Reset votes

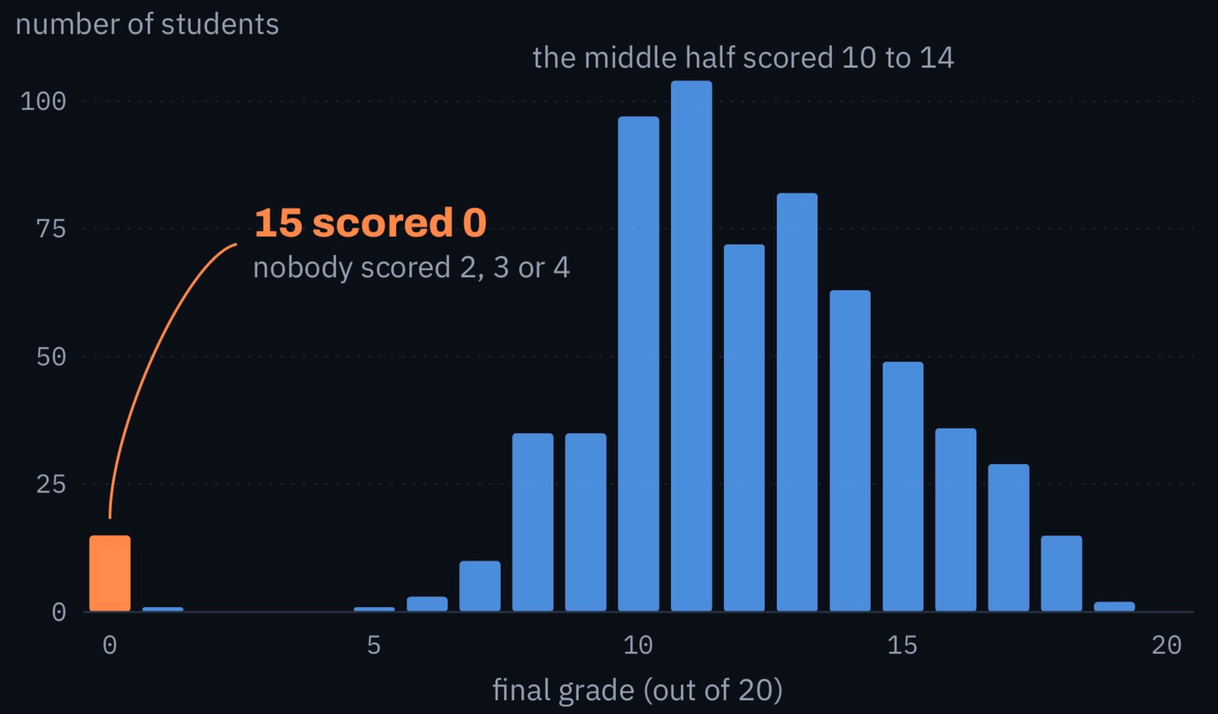
Data: Cortez (2008), UCI Student Performance, CC BY 4.0

649 students, two schools, one final grade out of 20

15

students scored **exactly 0**. We call them **zero-grade cases**: students who got 0 and probably left. Keep an eye on them.

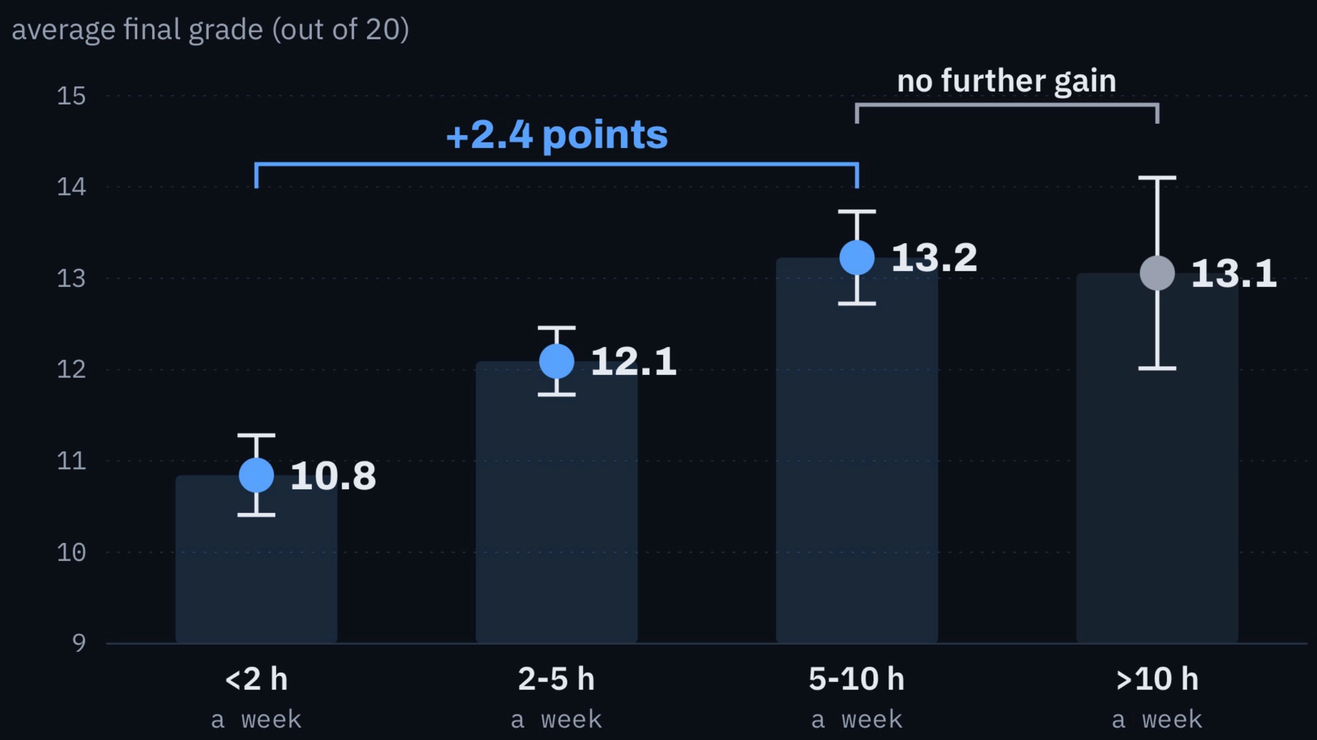
- Study time was a tick-box (<2, 2-5, 5-10, >10 h a week), not a stopwatch.
- We watched, we didn't experiment: these are patterns, not proof.



Stats details N TOOL USED Distributions Descriptive statistics

Maths file (395 students) kept aside as a second check

Do students who study more get better grades?



+2.4

points out of 20, from **under 2 h** to **5-10 h** a week. That is about **three-quarters** of the usual spread in grades (3.2 points).

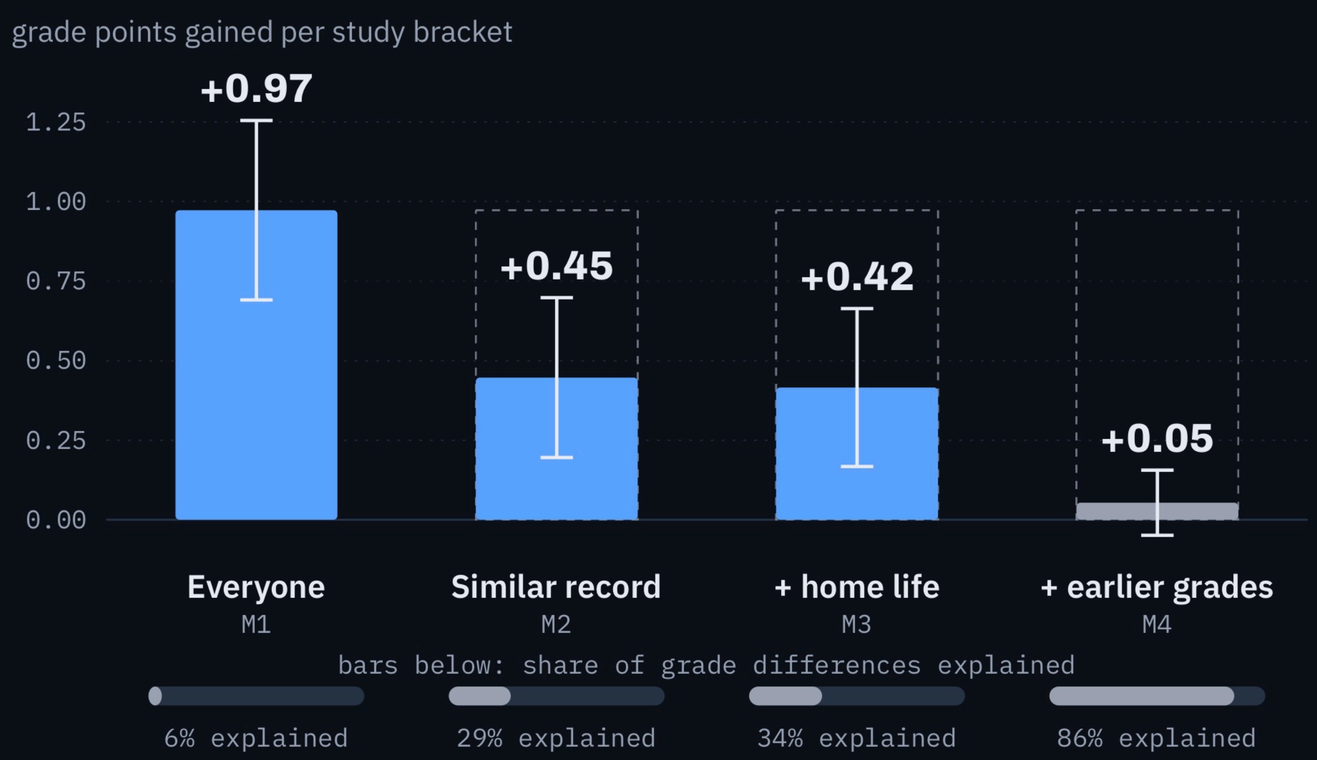
> Past 10 hours a week, no extra gain. Few students study that much, so treat this end with care.

Stats details N

TOOL USED One-way ANOVA · Tukey HSD · Kruskal-Wallis · MANOVA

Averages and 95% CIs computed live from the data. The grade axis starts at 9.

Is it the studying, or the kind of student who studies?



+0.05

per bracket, once we also know their earlier grades in this course

THINK OF IT LIKE

judging a fertiliser fairly: compare plants that got the same sun and water first.

Still real. Matched on record and home life, studying keeps a small, clear effect.

Earlier grades soak up the rest. But those grades were partly earned by studying, so this last step undersells effort.

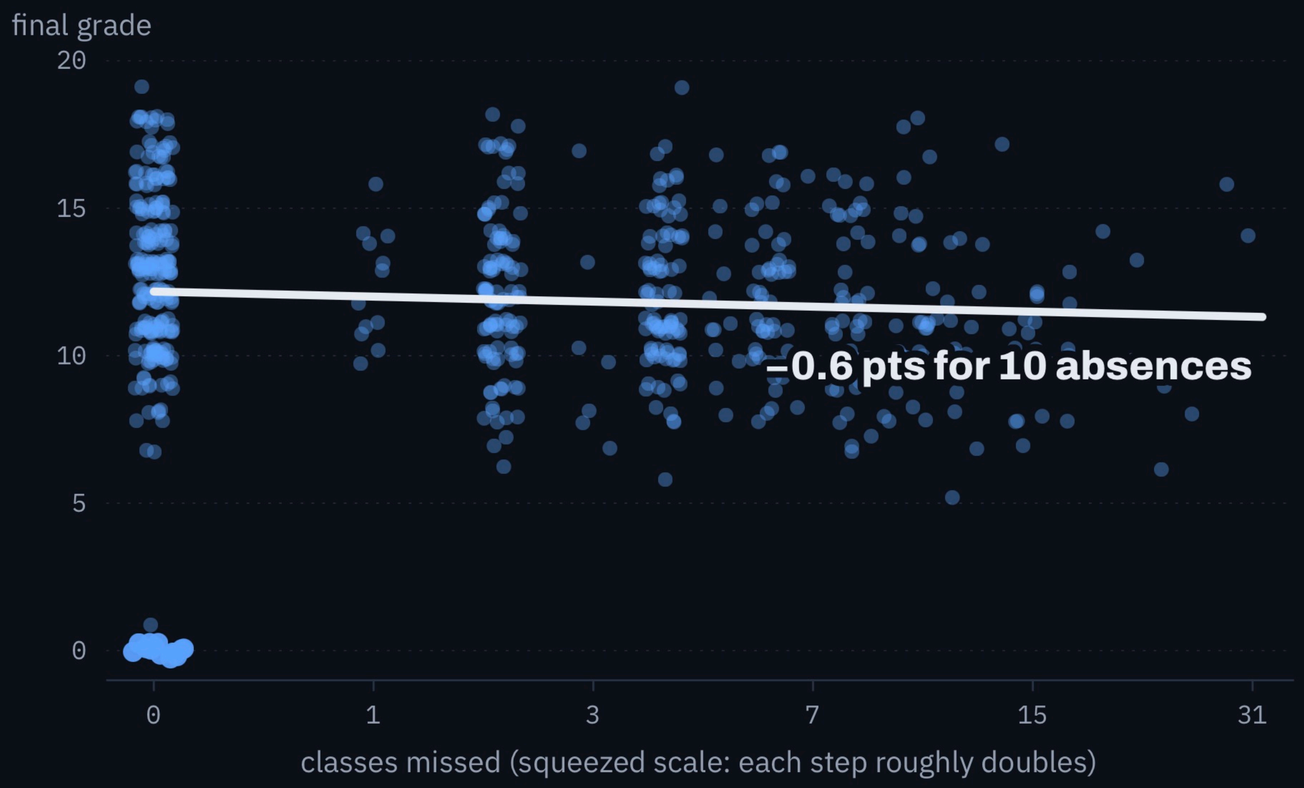
Stats details N

TOOL USED Multiple regression (OLS, HC3) · Dummy variables · ANCOVA

Whiskers: 95% CI. n = 649.

ACT 3 THE PERFECT ATTENDANCE PARADOX

Why does missing class seem not to matter?



Zero-grade cases: included · students · scored 0

WITH THEM
–0.6
points for 10 missed classes

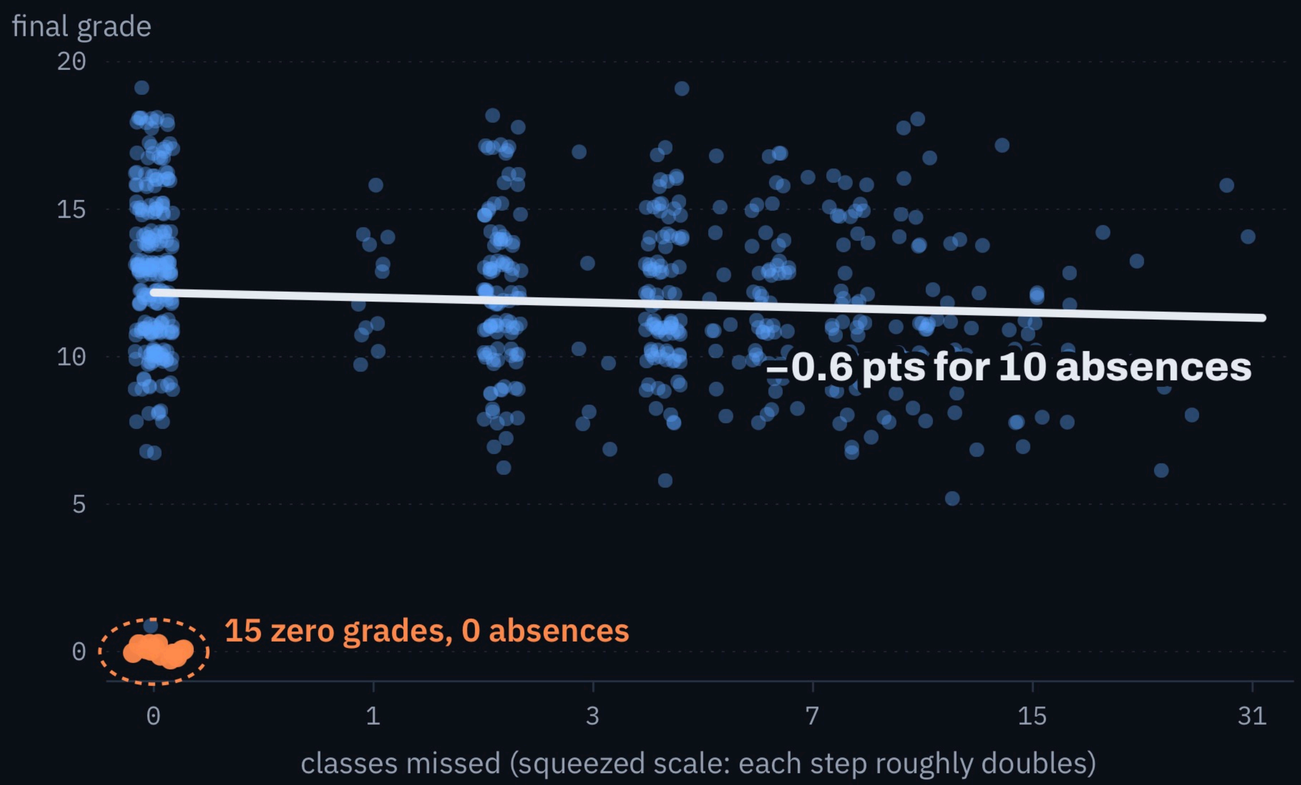
WITHOUT THEM
–1.4
points for 10 missed classes

Stats details N · TOOL USED · OLS with and without · Fisher's exact · Functional form: log(abs+1) · Breusch-Pagan

Points jittered. Lines fitted live. n = 649 / 634

ACT 3 THE PERFECT ATTENDANCE PARADOX

Why does missing class seem not to matter?



Zero-grade cases: included · students · scored 0

15 of 15

students who scored 0 had **zero recorded absences**. Among everyone, fewer than 4 in 10 did.

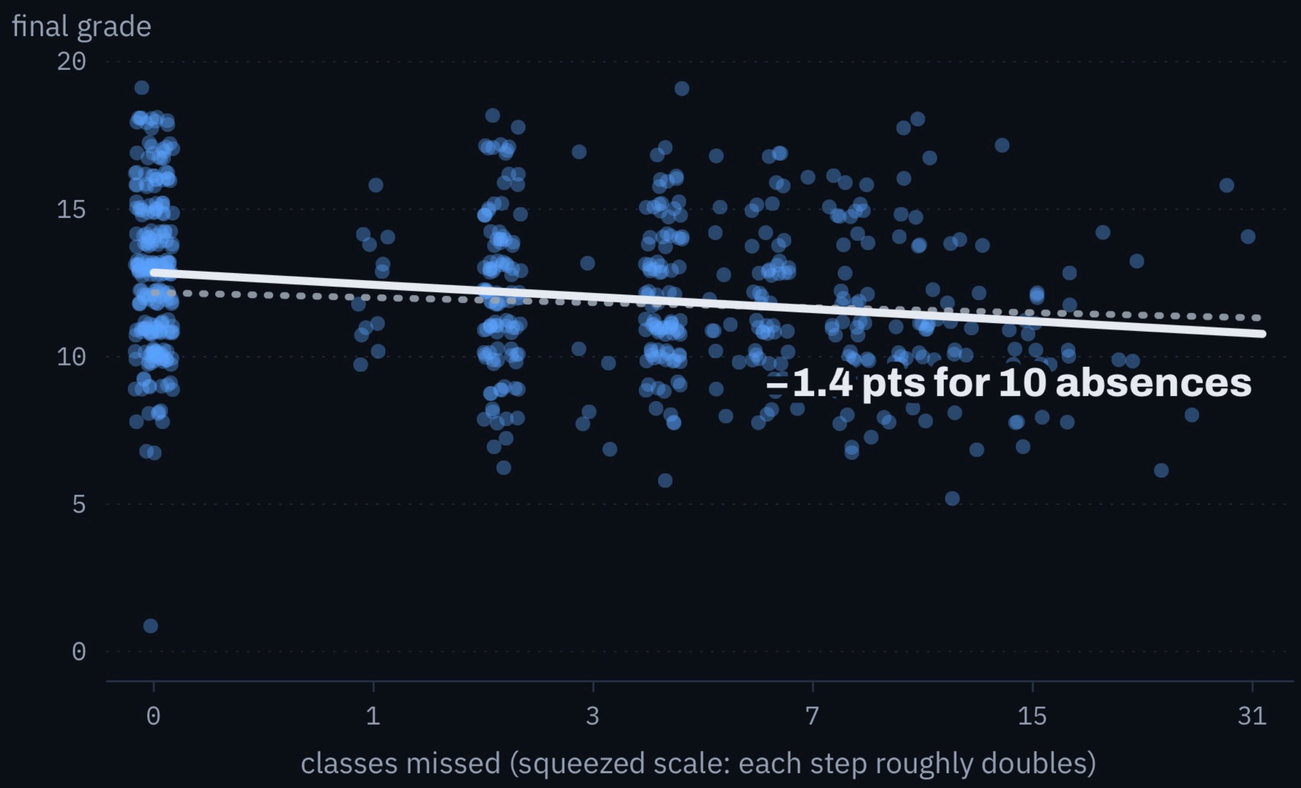
WITH THEM
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Stats details N · TOOL USED · OLS with and without · Fisher's exact · Functional form: log(abs+1) · Breusch-Pagan

Points jittered. Lines fitted live. n = 649 / 634

Why does missing class seem not to matter?



Zero-grade cases: excluded

students

scored 0

15 of 15

students who scored 0 had **zero recorded absences**. Among everyone, fewer than 4 in 10 did.

WITH THEM

−0.6

points for 10 missed classes

WITHOUT THEM

−1.4

points for 10 missed classes

Their grades were already sliding (about 7, then 4, then 0). We read them as **apparent dropouts**: a register can only mark you absent while you are still on it. That is our interpretation; the data never say why.

Stats details N

TOOL USED

OLS with and without

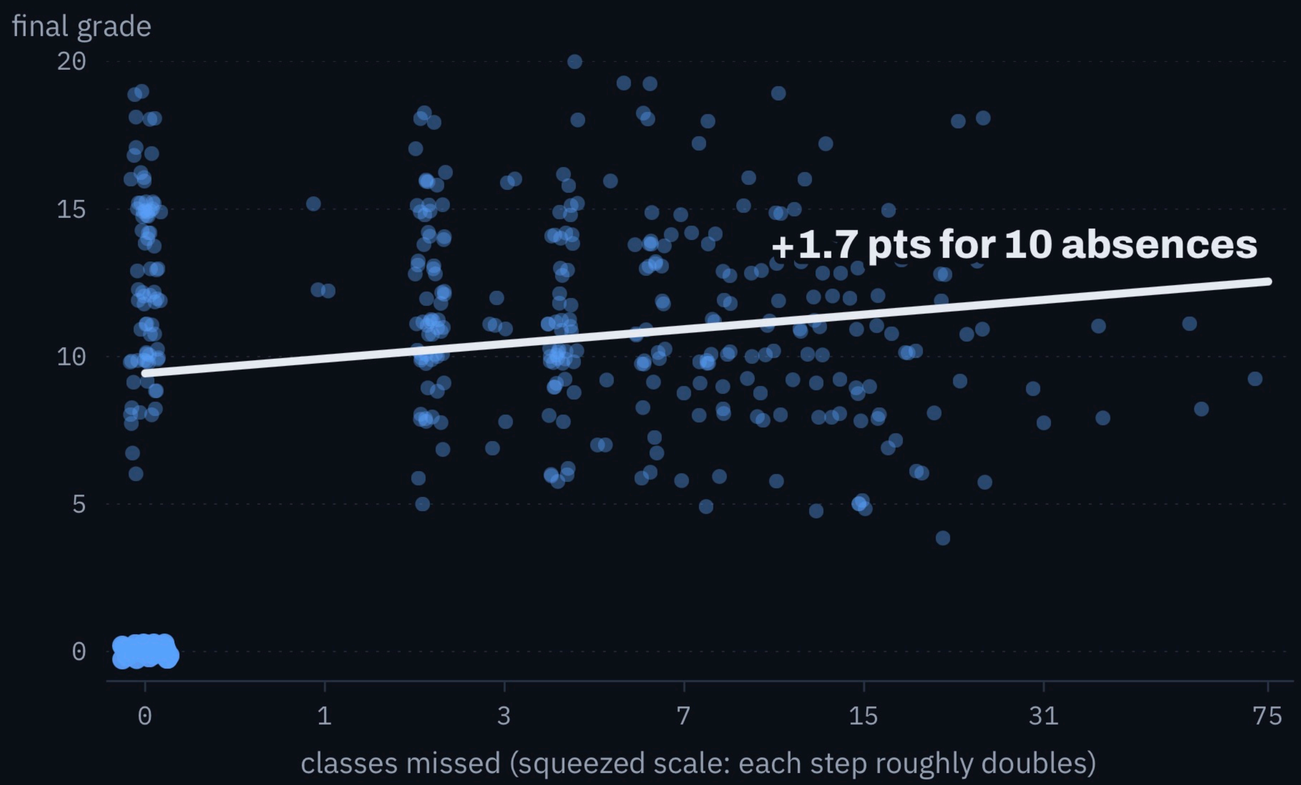
Fisher's exact

Functional form: log(abs+1)

Breusch-Pagan

Points jittered. Lines fitted live. n = 649 / 634

Same check, different subject: what about maths?



Zero-grade cases: included

WITH THEM

+1.7

points for 10 missed classes

WITHOUT THEM

−1.9

points for 10 missed classes

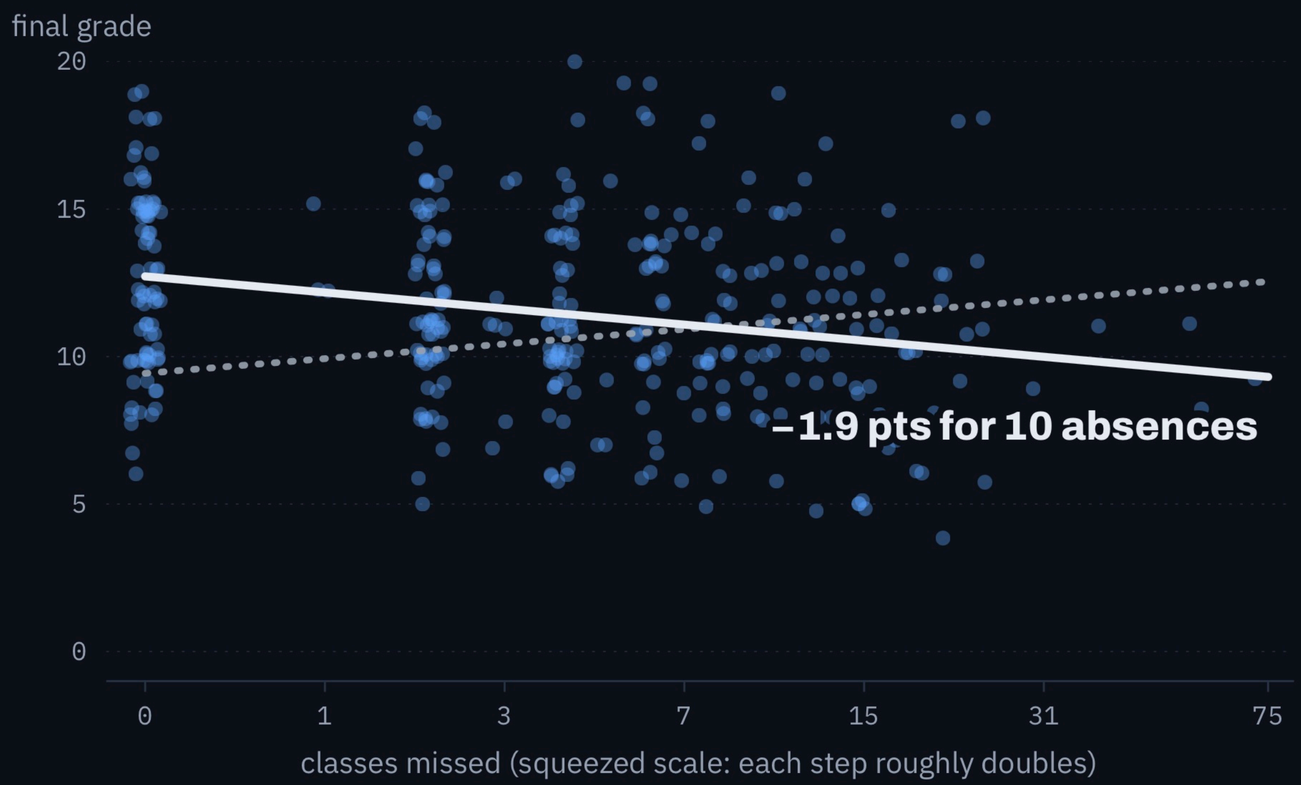
Leave the zeros in and skipping class seems to **help**. Take them out and the line flips: absences cost points.

One thing did **not** repeat: in Portuguese, 14 of the 15 zeros came from one school. In maths they were spread across both schools in line with school size, so we make no school claim.

Stats details N TOOL USED Replication on a second file OLS

Maths course, n = 395 / 357. Many students appear in both files.

Same check, different subject: what about maths?



Zero-grade cases: excluded

WITH THEM

+1.7

points for 10 missed classes

WITHOUT THEM

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points for 10 missed classes

Leave the zeros in and skipping class seems to **help**. Take them out and the line flips: absences cost points.

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Stats details N TOOL USED Replication on a second file OLS

Maths course, n = 395 / 357. Many students appear in both files.

So what does predict a final grade?

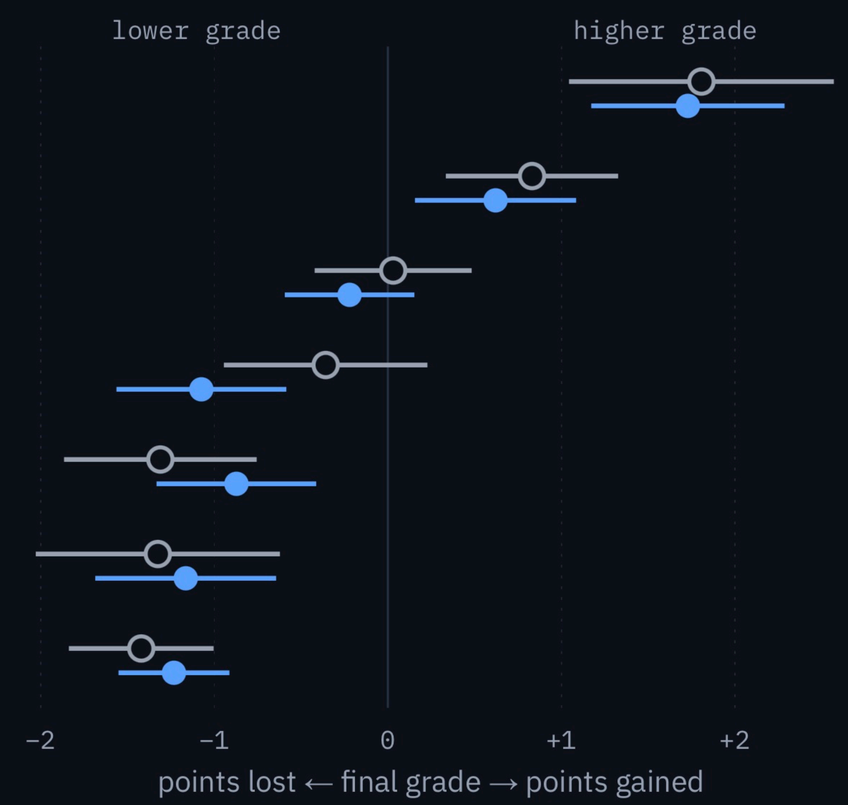
+3.5

points for students who **want higher education**. More than the usual spread in grades. Still about **+1.8** when we compare similar students.

WHY EXTRA SUPPORT LOOKS BAD

Like hospitals: the sickest patients get the most treatment. Support goes to students already struggling; it does not cause the low grade.

all 649 zero-grade cases removed



Stats details N

TOOL USED OLS with dummy variables | Welch t-test | Simultaneity (named, not solved)

Each row: points gained or lost, other things held equal (M3).

IN ONE LINE **Wanting higher education matters most; past failures hurt; family support shows no effect.**

YOUR TURN PREDICT, THEN WE REVEAL

Hands up: who does the model predict will score higher?

STUDENT A

Studies **>10 h** a week
1 past failure
12 absences
No extra school support

0 votes

STUDENT B

Studies **2-5 h** a week
0 past failures
2 absences
Gets extra school support

0 votes

Undo last | Reset votes | keys 1 / 2

Stats details N | TOOL USED | Prediction from M3 | Confidence vs prediction interval

Other characteristics at typical values. n = 649.

YOUR TURN PREDICT, THEN WE REVEAL

The model says A: 12.6, B: 12.2

STUDENT A

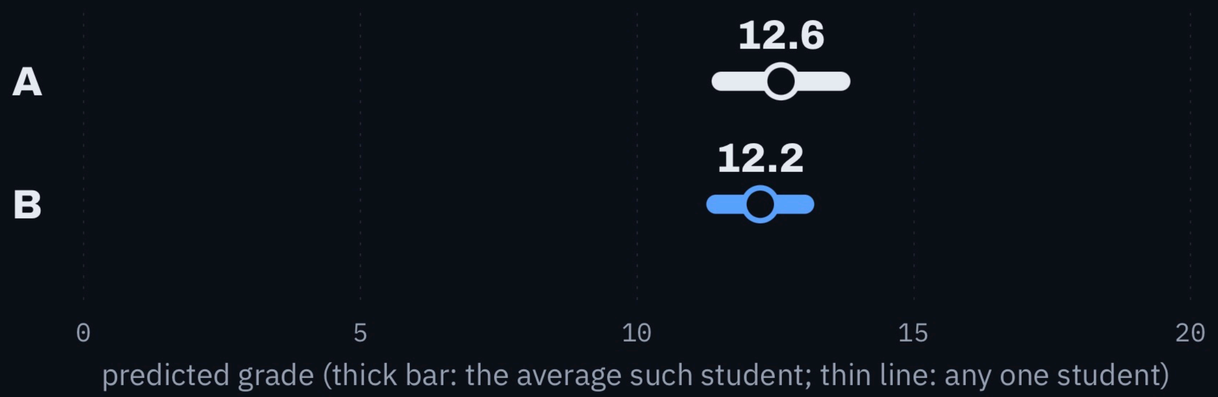
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No extra school support

0 votes

STUDENT B

Studies 2-5 h a week
0 past failures
2 absences
Gets extra school support

0 votes



predicted grade (thick bar: the average such student; thin line: any one student)

Undo last Reset votes keys 1 / 2

Stats details N TOOL USED Prediction from M3 Confidence vs prediction interval

Other characteristics at typical values. n = 649.

YOUR TURN PREDICT, THEN WE REVEAL

A model predicts the average, not the person

STUDENT A

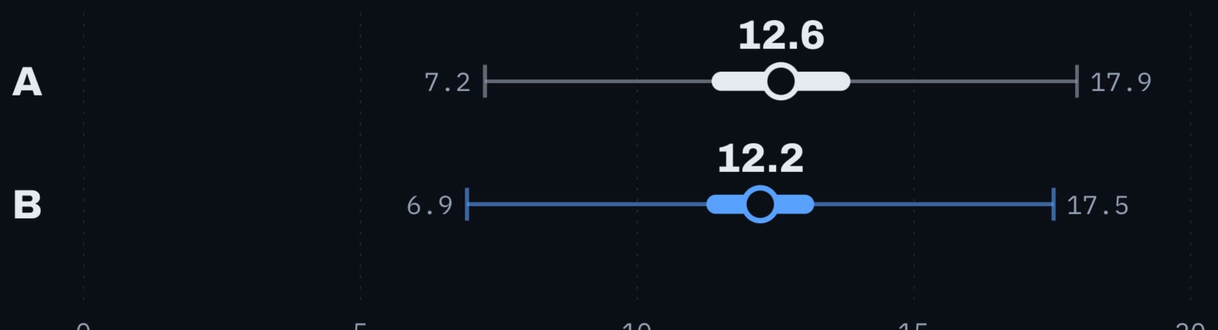
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Studies **2-5 h** a week
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2 absences
Gets extra school support

0 votes



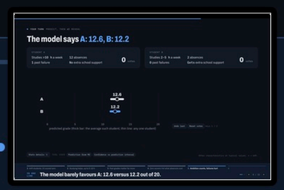
Real students like these could land anywhere from about 7 to 18.

Undo last | Reset votes | keys 1 / 2

Stats details N | TOOL USED | Prediction from M3 | Confidence vs prediction interval

Other characteristics at typical values. n = 649.

1 649 students; 15 scored zero | 2 Studying helps, up to 5-10 h | 3 Past grades matter far more | 4 Zero scorers hid what absences cost | 5 **Ambition counts, failures hurt** | 6



IN ONE LINE **A model predicts the average student, not any one person.**

YOUR TURN PREDICT, THEN WE REVEAL

The model's pick flips once you see who gets support

STUDENT A

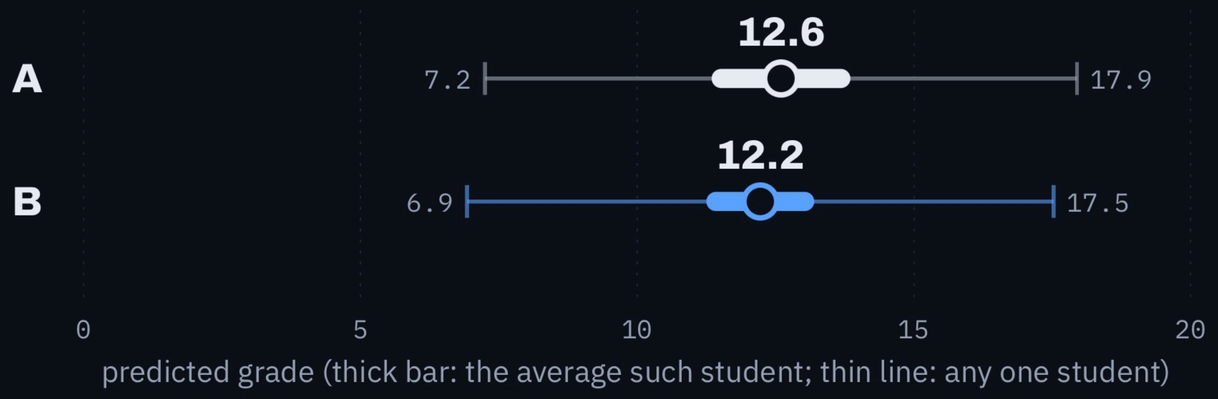
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Studies **2-5 h** a week
0 past failures
2 absences
Gets extra school support

0 votes



Real students like these could land **anywhere from about 7 to 18.**

The twist: A is ahead only because B gets extra support, which marks a struggling student (slide 7). Hold support equal and **B leads by 0.93**.

Undo last | Reset votes | keys 1 / 2

Stats details N | TOOL USED | Prediction from M3 | Confidence vs prediction interval

Other characteristics at typical values. n = 649.

Can we trust the model behind these numbers?

THE QUESTION WE ASKED	RESULT	WHAT WE DID
Are the predictors tangled together? VIF	PASS	Nothing needed
Is the scatter around the line even? Breusch-Pagan	WATCH	Used robust error bars (HC3) everywhere
Is a straight-line model the right shape? RESET	WATCH	Reported openly; grades capped at 0-20 are a likely cause
Should absences be squeezed (log scale)? AIC, squared term	PASS	Log scale; same answer either way
Do the errors follow a bell curve? Jarque-Bera	PASS	Once the zero-grade cases are set aside
Does any one student drive the results? Cook's distance	PASS	No single point dominates
Does row order matter? Durbin-Watson	N/A	No: one snapshot of students, not a time series

Stats details N TOOL USED VIF Breusch-Pagan / HC3 RESET Functional form

Results shown with zero-grade cases removed; both versions in Stats details.

Effort shows up. History dominates. Check your zeros.

THE STORY SO FAR, ALL SIX PARTS

1. Real grades from 649 students, and 15 of them scored zero.
   slide 2
2. Studying more helps, up to about 5-10 hours a week.
   slide 3
3. Studying helps, but your past grades matter far more.
   slide 4
4. Every student who scored zero had perfect attendance; without them, absences hurt. Maths shows it too.
   slides 5-6
5. Wanting higher education matters most, past failures hurt, and a model predicts averages, not people.
   slides 7-8
6. The model mostly passes its checks, but these are patterns, not proof.
   slides 9, 11

Stats details N

For you: your last grade predicts your next, so change course early.

What can't this study prove?

1 **Patterns, not proof.**
Students choose how much to study, so keen students may differ in ways we can't see.

2 **Earlier grades blur the effort effect.**
Those grades were partly earned by studying, so the last model undersells effort.

3 **"Apparent dropouts" is our reading.**
The data never say why a grade is 0, so every key result is shown with and without them.

4 **Rough measures.**
Study time is four tick-boxes, and grades are capped at 0-20.

5 **One place, one year.**
Two Portuguese schools in 2005-06, with few students in the top study group.

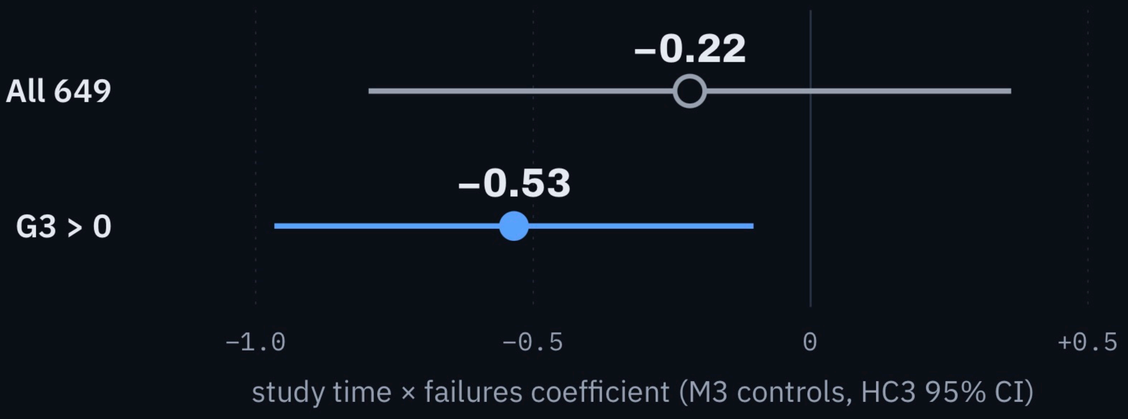
Thank you. Questions?

Cortez, P. (2008). Student Performance [Dataset]. UCI Machine Learning Repository. doi.org/10.24432/C5TG7T. CC BY 4.0.
Cortez, P. & Silva, A. (2008). Using data mining to predict secondary school student performance. FUBUTEC 2008, 5-12.

Stats details N
TOOL USED
Endogeneity / simultaneity
Model specification
Backup slides follow (→)

IN ONE LINE **These are patterns, not proof that studying causes better grades.**

Does studying help repeaters more? Only when the zero-grade cases are removed, so we call it fragile



Study time × past failures, added to M3.

ALL 649 −0.22, 95% CI −0.80 to 0.36, p = 0.46

G3 > 0 −0.53, 95% CI −0.97 to −0.10, p = 0.015

Negative means the study-time slope is flatter for students with past failures. Few repeaters study 5 h or more, so we keep this out of the main story.

TOOL USED OLS interaction term

results/regression_comparison.csv

IN ONE LINE **Studying may help repeaters less, but this result is fragile.**

The ladder M1 to M4, with and without the zero-grade cases

MODEL	STUDY TIME B [95% CI]	P (LINEAR)	P (3 DUMMIES, JOINT)	LOG(ABS+1) B (P)	R^2	ADJ. R^2	N
ALL STUDENTS							
M1 study time	+0.97 [+0.69, +1.25]	< 0.001	< 0.001	–	0.062	0.061	649
M2 + academic/demographic	+0.45 [+0.20, +0.70]	< 0.001	0.002	−0.16 (0.185)	0.287	0.278	649
M3 + family/social	+0.42 [+0.17, +0.66]	0.001	0.005	−0.15 (0.231)	0.338	0.317	649
M4 + G1, G2	+0.05 [−0.05, +0.16]	0.307	0.633	+0.16 (0.022)	0.856	0.851	649
ZERO-GRADE CASES REMOVED (G3 > 0)							
M1 study time	+0.80 [+0.54, +1.05]	< 0.001	< 0.001	–	0.060	0.059	634
M2 + academic/demographic	+0.33 [+0.10, +0.57]	0.005	0.032	−0.48 (< 0.001)	0.302	0.293	634
M3 + family/social	+0.31 [+0.08, +0.54]	0.009	0.063	−0.45 (< 0.001)	0.369	0.348	634
M4 + G1, G2	+0.01 [−0.07, +0.10]	0.747	0.701	−0.01 (0.881)	0.888	0.884	634

TOOL USED OLS, HC3 | Dummy variables | MANCOVA (+ failures): Pillai 0.065, p = 2.6e-6

M2 adds failures, log(abs+1), sex, age, school, address, higher. M3 adds Medu, Fedu, goout, Dalc, Walc, freetime, health, internet, schoolsup, famsup, paid, romantic.

IN ONE LINE **The full table behind the shrinking study-time effect.**

← Backup A3 / A5 →

However we treat the zero-grade cases, absences turn negative and the study ladder holds

TREATMENT OF THE 15 CASES	N	M3 LOG(ABS+1) (P)	M3 STUDY TIME (P)	M4 LOG(ABS+1) (P)	M4 STUDY TIME (P)
Keep as ordinary observations	649	−0.15 (0.23)	+0.42 (0.001)	+0.16 (0.022)	+0.05 (0.31)
(a) Drop them (G3 > 0)	634	−0.45 (< 0.001)	+0.31 (0.009)	−0.01 (0.88)	+0.01 (0.75)
(b) Zero-grade dummy	649	−0.47 (< 0.001)	+0.31 (0.008)	−0.03 (0.63)	+0.03 (0.56)
(c) Restrict to G2 > 0	642	−0.28 (0.019)	+0.40 (0.001)	+0.15 (0.051)	+0.05 (0.29)

TOOL USED Sensitivity analysis Dummy variables

results/zero_grade_robustness.csv

Every number in this deck traces to the notebooks and results files

Data. Cortez, P. (2008). *Student Performance* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5TG7T. Licensed CC BY 4.0.

Original study. Cortez, P., & Silva, A. (2008). Using data mining to predict secondary school student performance. In *Proceedings of the 5th Future Business Technology Conference (FUBUTEC 2008)*, pp. 5-12. EUROSIS.

Methods. Python: pandas, scipy, statsmodels. OLS with HC3 robust standard errors throughout. Notebooks 00-07 in the project repository; numbers in `results/*.csv`.

Live charts. The scatter plots, group means and histogram on these slides are recomputed in the browser from the embedded data (study time, failures, sex, absences, G1, G2, G3, zero_grade) and checked against the notebook values on load. "Usual spread" = the standard deviation of the final grade (3.23), also computed live. Open the browser console to see the checks.

In-browser checks against notebooks 02-06: 49 of 49 passed.

TOOL USED Reproducible pipeline

IN ONE LINE **Every number here traces back to our notebooks and results files.**